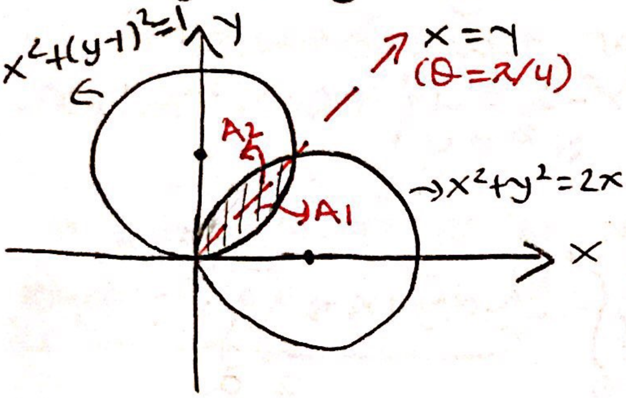
EXERCISES

1) Evaluate the area of the region that is bounded by the curves $x^2+y^2=2x$ and $x^2+y^2=2y$.

$$x^2+y^2=2x \Rightarrow (x-1)^2+y^2=1$$
$$x^2+y^2=2y \Rightarrow x^2+(y-1)^2=1$$

intersection $x=y$



$$A=\iint_D dx\,dy$$

with polar transformation;

$$A=\iint_D r\,dr\,d\theta$$

$$x^2+y^2=2x \Rightarrow r^2=2r\cos\theta \Rightarrow r=2\cos\theta \quad (\text{and } r=0)$$
$$x^2+y^2=2y \Rightarrow r^2=2r\sin\theta \Rightarrow r=2\sin\theta \quad (\text{and } r=0)$$

$$A=A_1+A_2$$

$$A=\int_0^{\pi/4}\int_0^{2\sin\theta} r\,dr\,d\theta+\int_{\pi/4}^{\pi/2}\int_0^{2\cos\theta} r\,dr\,d\theta$$

$$A=\int_0^{\pi/4}\frac{r^2}{2}\Big|_0^{2\sin\theta}d\theta+\int_{\pi/4}^{\pi/2}\frac{r^2}{2}\Big|_0^{2\cos\theta}d\theta$$

$$A=\frac{1}{2}\int_0^{\pi/4}4\sin^2\theta\,d\theta+\frac{1}{2}\int_{\pi/4}^{\pi/2}4\cos^2\theta\,d\theta$$

$$A=2\int_0^{\pi/4}\left[\frac{1-\cos 2\theta}{2}\right]d\theta+2\int_{\pi/4}^{\pi/2}\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$A=\left[\theta-\frac{1}{2}\sin 2\theta\right]\Big|_0^{\pi/4}+\left[\theta+\frac{1}{2}\sin 2\theta\right]\Big|_{\pi/4}^{\pi/2}$$

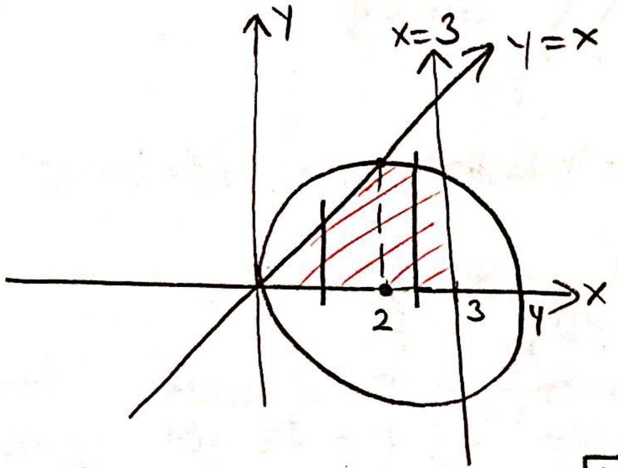
$$A=\cancel{\frac{\pi}{4}}-\frac{1}{2}+\frac{\pi}{2}+0-\cancel{\frac{\pi}{4}}-\frac{1}{2} \Rightarrow A=\frac{\pi}{2}-1$$

2) $D: \begin{cases} x^2+y^2-4x=0 \\ y=0 \ , \ y=x \\ x=3 \end{cases}$ and $I=\iint_D 2y\,dx\,dy$

are given.

a) Evaluate I with respect to x.
b) Evaluate I " " " y.
c) Evaluate I by using polar transformation.
d) Evaluate I by using $\left.\begin{array}{l} x-2=u \\ y^2=v \end{array}\right\}$ transformation.

a)



Region is not smooth wrt-x

$$I=\int_0^2\int_0^x 2y\,dy\,dx+\int_2^3\int_0^{\sqrt{4x-x^2}} 2y\,dy\,dx$$

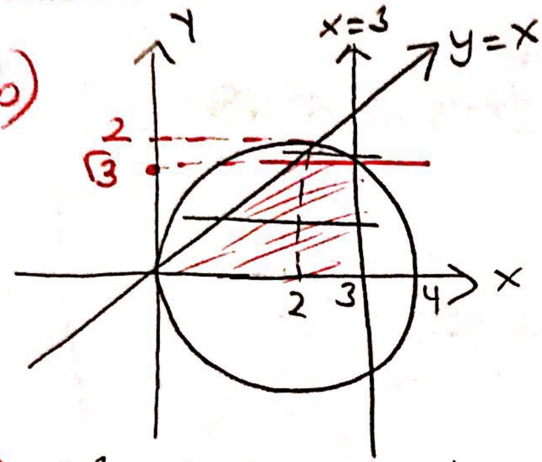
$$I=\int_0^2 y^2\Big|_0^x dx+\int_2^3 y^2\Big|_0^{\sqrt{4x-x^2}} dx$$

$$I=\int_0^2 x^2dx+\int_2^3(4x-x^2)dx$$

$$I=\frac{x^3}{3}\Big|_0^2+\left(2x^2-\frac{x^3}{3}\right)\Big|_2^3$$

$$I=\frac{8}{3}+\underbrace{(18-9)}_{9}-\left(8-\frac{8}{3}\right)$$

$$I=\frac{16}{3}+1=\frac{19}{3}$$

b) 

Region is not smooth with respect to y-axis.

$$I=\int_0^{\sqrt{3}}\int_y^3 2y\,dx\,dy+\int_{\sqrt{3}}^2\int_y^{2+\sqrt{4-y^2}} 2y\,dx\,dy$$

(*) $x^2+y^2=4x$

$(x-2)^2+y^2=4$

$(x-2)^2=4-y^2$

$x-2=\mp\sqrt{4-y^2}$

$x=2\mp\sqrt{4-y^2}$

$$I=\int_0^{\sqrt{3}} 2yx\Big|_y^3\,dy+\int_{\sqrt{3}}^2 2yx\Big|_y^{2+\sqrt{4-y^2}}\,dy$$

$$I=\int_0^{\sqrt{3}} 2y(3-y)\,dy+\int_{\sqrt{3}}^2 2y(2+\sqrt{4-y^2}-y)\,dy$$

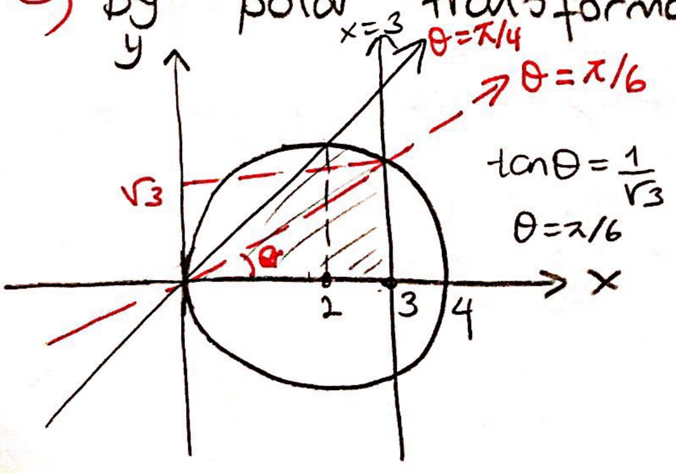
$$I=\int_0^{\sqrt{3}}(6y-2y^2)\,dy+\int_{\sqrt{3}}^2(4y+2y\sqrt{4-y^2}-2y^2)\,dy$$

$$I=\left(3y^2-\frac{2}{3}y^3\right)\Big|_0^{\sqrt{3}}+\left(2y^2-\frac{2}{3}y^3\right)\Big|_{\sqrt{3}}^2+2\int_{\sqrt{3}}^2 y\sqrt{4-y^2}\,dy$$

$$I=9-\frac{2}{3}.3\sqrt{3}+\left(8-\frac{2}{3}.8\right)-\left(6-\frac{2}{3}.3\sqrt{3}\right)-\frac{2}{3}(4-y^2)^{3/2}\Big|_{\sqrt{3}}^2$$

$$I=\frac{19}{3}$$

c) By polar transformation;



$\tan\theta=\frac{1}{\sqrt{3}}$

$\theta=\pi/6$

$$I=\int_0^{\pi/6}\int_0^{3/\cos\theta} 2r\sin\theta.r\,dr\,d\theta+$$

$$\int_{\pi/6}^{\pi/4}\int_0^{4\cos\theta} 2r\sin\theta.r\,dr\,d\theta$$

$$I = \int_0^{\pi/6}\int_0^{3/\cos\theta} 2r^2\sin\theta\, dr\, d\theta + \int_{\pi/6}^{\pi/4}\int_0^{4\cos\theta} 2r^2\sin\theta\, dr\, d\theta$$

$$I = \frac{2}{3}\int_0^{\pi/6} r^3\sin\theta \Big|_0^{3/\cos\theta} d\theta + \frac{2}{3}\int_{\pi/6}^{\pi/4} r^3\sin\theta\Big|_0^{4\cos\theta} d\theta$$

$$I = \frac{2}{3}\int_0^{\pi/6}\frac{27}{\cos^3\theta}\cdot\sin\theta\, d\theta + \frac{2}{3}\int_{\pi/6}^{\pi/4} 64\cos^3\theta\sin\theta\, d\theta$$

$\cos\theta = t$

$-\sin\theta\, d\theta = dt$

$$I = -18\int_1^{\sqrt{3}/2}\frac{1}{t^3}\,dt - \frac{128}{3}\int_{\sqrt{3}/2}^{\sqrt{2}/2} t^3\, dt$$

$$I = -\frac{18}{-2}\cdot t^{-2}\Big|_1^{\sqrt{3}/2} - \frac{128}{3}\cdot\frac{t^4}{4}\Big|_{\sqrt{3}/2}^{\sqrt{2}/2}$$

$$I = 9\left(\frac{4}{3} - 1\right) + \frac{32}{3}\left(\frac{9}{16} - \frac{1}{4}\right)$$

$$I = \frac{19}{3}$$

d) $x = u + 2$

$y = \sqrt{v}$

D':

* $x^2 + y^2 = 4x$

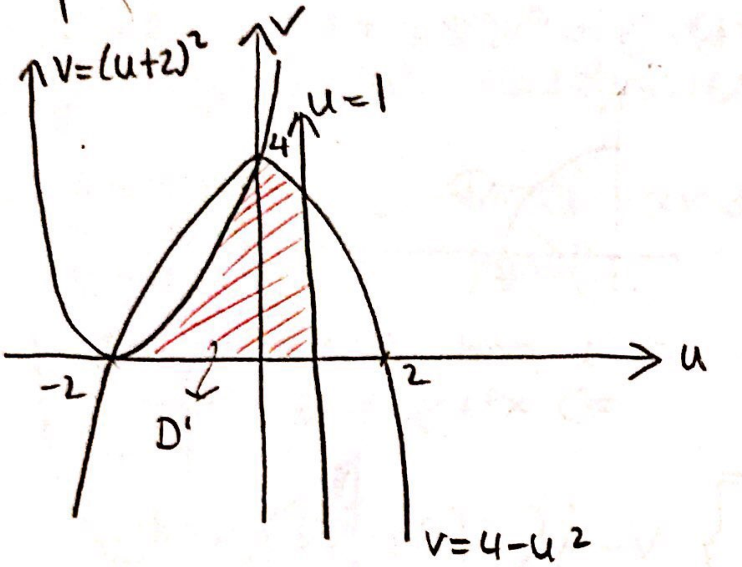
  $(u+2)^2 + v = 4(u+2) \Rightarrow v = 4 - u^2$

* $y = 0 \Rightarrow v = 0$

* $y = x \Rightarrow v = (u+2)^2$

* $x = 3 \Rightarrow u = 1$

$$D' : \left\{ v = 4-u^2 \;,\; v = 0 \;,\; v = (u+2)^2 \;,\; u = 1 \right.$$



$$J = \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & \frac{1}{2\sqrt{v}} \end{vmatrix} = \frac{1}{2\sqrt{v}}$$

$$I = \int_{-2}^{0}\int_{0}^{(u+2)^2} \cancel{2\sqrt{v}} \cdot \frac{1}{\cancel{2\sqrt{v}}} \, dv\,du + \int_{0}^{1}\int_{0}^{4-u^2} \cancel{2\sqrt{v}} \cdot \frac{1}{\cancel{2\sqrt{v}}} \, dv\,du$$

$$I = \int_{-2}^{0} (u+2)^2 du + \int_{0}^{1} (4-u^2) du$$

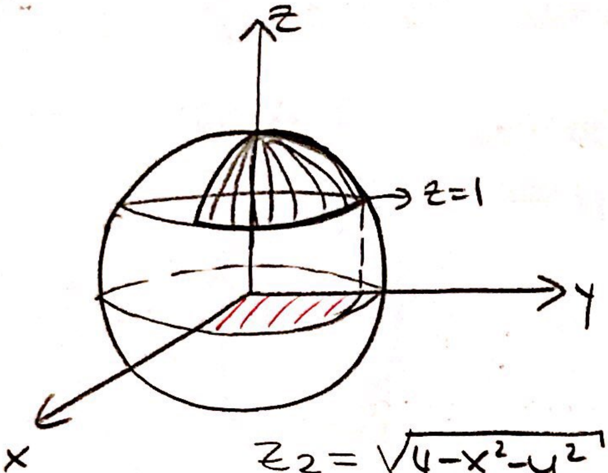
$$I = \int_{-2}^{0} (u^2+4u+4) du + \int_{0}^{1} (4-u^2) du$$

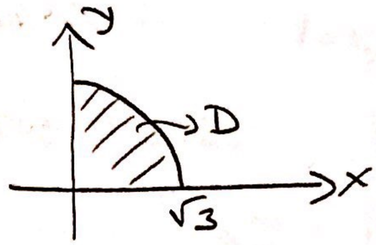
$$I = \frac{u^3}{3} + 2u^2 + 4u \Big|_{-2}^{0} + \left(4u - \frac{u^3}{3}\right) \Big|_{0}^{1}$$

$$I = 0 - \left(-\frac{8}{3} + \cancel{8} - \cancel{8}\right) + 4 - \frac{1}{3} - 0 = 4 + \frac{7}{3}$$

$$I = \frac{19}{3}$$

3) Evaluate the volume of the solid bounded by $z=\sqrt{4-x^2-y^2}$ and $x \geq 0, y \geq 0, z \geq 1$.





$z=1$ and $z^2+x^2+y^2=4$
$\Rightarrow x^2+y^2=3$

$$\left.\begin{array}{l} z_2=\sqrt{4-x^2-y^2} \\ z_1=1 \end{array}\right\} V=\iint_D (z_2-z_1)\,dx\,dy$$

$$V=\iint_D \left(\sqrt{4-x^2-y^2}-1\right)dx\,dy$$, by polar transform.

$$V=\int_0^{\pi/2}\int_0^{\sqrt{3}}\left(\sqrt{4-r^2}-1\right)r\,dr\,d\theta=\int_0^{\pi/2}\int_0^{\sqrt{3}}\left(r\sqrt{4-r^2}-r\right)dr\,d\theta$$

$$V=\int_0^{\pi/2}\int_0^{\sqrt{3}} r\sqrt{4-r^2}\,dr\,d\theta-\int_0^{\pi/2}\int_0^{\sqrt{3}} r\,dr\,d\theta$$

$4-r^2=t^2$
$-\cancel{2}r\,dr=\cancel{2}t\,dt$

$$V=\int_0^{\pi/2}\int_2^{1} t(-t\,dt)\,d\theta-\frac{1}{2}\int_0^{\pi/2} r^2\Big|_0^{\sqrt{3}}\,d\theta$$

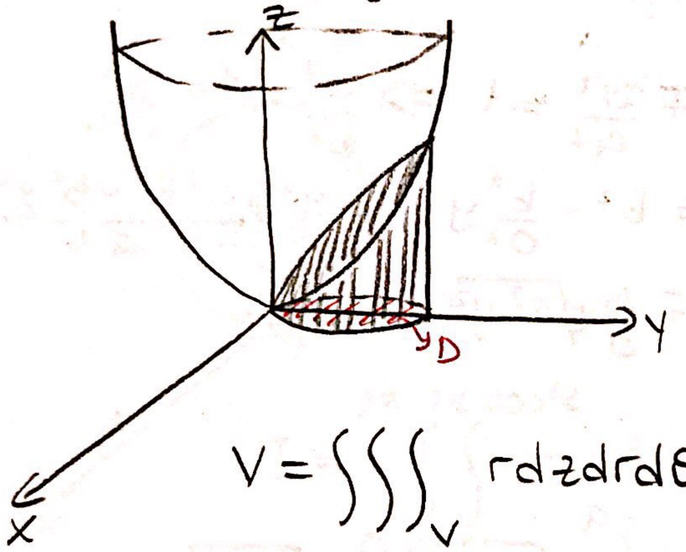
$$V=\frac{1}{3}\int_0^{\pi/2} t^3\Big|_1^2\,d\theta-\frac{1}{2}\int_0^{\pi/2} 3\,d\theta$$

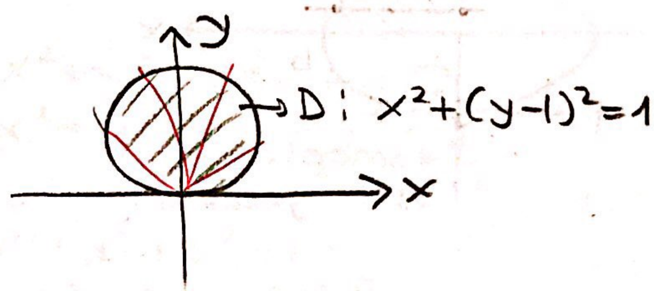
$$V=\frac{1}{3}\int_0^{\pi/2}(8-1)\,d\theta-\frac{3}{2}\cdot\frac{\pi}{2}=\frac{7}{3}\cdot\frac{\pi}{2}-\frac{3\pi}{4}$$

$$V=\frac{5\pi}{12}$$

4) Evaluate the volume bounded by the surfaces $4z = x^2+y^2$, $x^2+y^2 = 2y$ and $z=0$ by using cylindrical coordinates.

* $\left.\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ z &= z \end{aligned}\right\}$ cylindrical coordinates transf.





$$x^2+y^2 = 2y \Rightarrow r = 2\sin\theta$$

$$V = \iiint_V r\,dz\,dr\,d\theta \qquad \begin{aligned} z_1 &= 0 \\ z_2 &= \frac{x^2+y^2}{4} = \frac{r^2}{4} \end{aligned}$$

$$V = \int_0^{\pi}\int_0^{2\sin\theta}\int_{z_1=0}^{z_2=r^2/4} r\,dz\,dr\,d\theta = \int_0^{\pi}\int_0^{2\sin\theta} \frac{r^3}{4}\,dr\,d\theta$$

$$V = \int_0^{\pi} \frac{r^4}{16}\Bigg|_0^{2\sin\theta} d\theta = \frac{1}{\cancel{16}}\int_0^{\pi} \cancel{2^4}\sin^4\theta\,d\theta = \int_0^{\pi}\left[\frac{1-\cos 2\theta}{2}\right]^2 d\theta$$

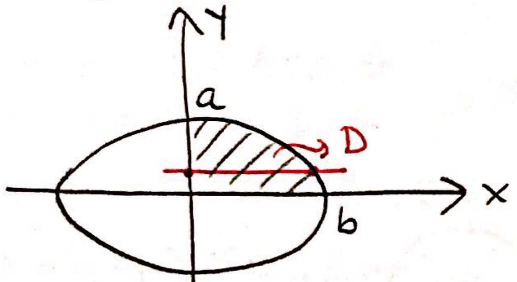
$$V = \frac{1}{4}\int_0^{\pi}(1 - 2\cos 2\theta + \cos^2 2\theta)\,d\theta$$

$$V = \frac{1}{4}\left[\theta - \sin 2\theta\right]\Big|_0^{\pi} + \frac{1}{4}\int_0^{\pi}\frac{(1+\cos 4\theta)}{2}\,d\theta$$

$$V = \frac{1}{4}[\pi - 0] + \frac{1}{8}\left[\theta + \frac{1}{4}\sin 4\theta\right]\Big|_0^{\pi}$$

$$V = \frac{\pi}{4} + \frac{1}{8}\cdot[\pi - 0] = \frac{\pi}{4} + \frac{\pi}{8} \Rightarrow V = \frac{3\pi}{8}$$

5) D: $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$, $x \geq 0$, $y \geq 0$ and $I=\iint_D \sqrt{a^2-y^2}\,dx\,dy$ are given. Evalute the integral I.



*smooth wrt y-axis.

$$I=\iint_D \sqrt{a^2-y^2}\,dx\,dy$$

$$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1 \Rightarrow \frac{x^2}{b^2}=1-\frac{y^2}{a^2}$$

$$x^2=b^2-\frac{b^2}{a^2}y^2 \Rightarrow x^2=\frac{a^2b^2-b^2y^2}{a^2}$$

$$x=\mp\frac{b}{a}\sqrt{a^2-y^2}$$

$$I=\int_0^a\int_0^{\frac{b}{a}\sqrt{a^2-y^2}} \sqrt{a^2-y^2}\,dx\,dy$$

$$I=\int_0^a \sqrt{a^2-y^2}\,.\,x\Big|_0^{\frac{b}{a}\sqrt{a^2-y^2}} dy=\int_0^a \sqrt{a^2-y^2}\cdot\frac{b}{a}\sqrt{a^2-y^2}\,dy$$

$$I=\frac{b}{a}\int_0^a (a^2-y^2)\,dy=\frac{b}{a}\left[a^2y-\frac{1}{3}y^3\right]\Big|_0^a$$

$$I=\frac{b}{a}\left[a^3-\frac{1}{3}a^3-0\right]=\frac{b}{a}\cdot\frac{2a^3}{3} \Rightarrow I=\frac{2}{3}a^2b$$